

Corrected quadratic payments for a fixed CVaR surrogate

Abstract

We examine the combination of a quadratic resource-allocation mechanism and an empirical conditional value-at-risk (CVaR) oracle. The mechanism's strict matrix certificate is incompatible with its implementation equalities, and its printed payment can violate individual rationality. An adjustment depending only on opponents' messages repairs equilibrium transfers. Under additional curvature and multiplier-existence assumptions, a weighted monotonicity argument implements the unique optimum of a fixed expected empirical CVaR surrogate, with exact budget balance and surrogate individual rationality. The oscillation of the surrogate error bounds population-risk regret, participation shortfall and welfare loss. Contracting only the loss queries preserves the zero outside action and local feasibility. A standard nested stochastic proximal-point construction gives mean-square allocation convergence to this fixed surrogate optimum, without requiring unique prices. These conclusions neither validate the original best-response learner nor establish exact population-risk learning.

1 The question and its scope

Garjani, Shokri and Kebriaei [3], called Q below, propose quadratic payments for strategic allocation with capacity constraints. Wang, Huang, Xu and Johansson [4], called P, develop a cooperative distributed CVaR method using loss values. P establishes consensus and convergence to a fixed expected empirical smoothed objective under its fixed-batch assumptions. Q's implementation and dynamic guarantees are conditional on certificates and assumptions that cannot simply be carried into that setting.

Our contribution is a correction and conditional combination of these constructions. We give a direct implementation proof for Q's explicit payment after an incentive-preserving transfer adjustment, and quantify how P's statistical surrogate changes economic guarantees. The new content is the specific repair and its combination with the economic error bounds. The value estimator, elementary smoothing construction and generic proximal-point convergence principle are not claimed as new. Throughout, implementation means Nash implementation in the message game; no truthful-reporting or dominant-strategy assertion is made.

2 Payments and implementation

2.1 Model and the printed payment

There are $N \geq 2$ agents and r resources. Agent n chooses $s_n = (x_n, p_n) \in X_n \times \mathbb{R}_+^r$, where X_n is compact and convex. Capacity is $c \in \mathbb{R}_+^r$. The planner minimises $\sum_n f_n(x_n)$ over

$$\mathcal{A} = \{x \in \prod_n X_n : \sum_n x_n \leq c\}.$$

The normal cone convention is $N_X(x) = \{v : \langle v, y - x \rangle \leq 0 \text{ for all } y \in X\}$. A planner Karush–Kuhn–Tucker (KKT) pair (x^*, λ) satisfies

$$0 \in \partial f_n(x_n^*) + \lambda + N_{X_n}(x_n^*), \quad \lambda \geq 0, \quad x^* \in \mathcal{A}, \quad \lambda^\top \left(\sum_n x_n^* - c \right) = 0. \quad (1)$$

The game cost is $f_n(x_n) + t_n(s)$; the coupling constraint is imposed at equilibrium, not on unilateral message choices.

Write $\gamma > 0$ for Q's payment scale, and define $P_{-n} = \sum_{j \neq n} p_j$, $X_{-n} = \sum_{j \neq n} x_j$, $S_{-n} = \sum_{j \neq n} \|p_j\|^2$, and $D_{-n} = \sum_{j \neq n} p_j^\top x_j$. The printed payment in Q, Section III, equation labelled **eq40** in its source, is

$$\begin{aligned} t_n^Q(s) = \gamma & \left[\frac{1}{2} \|p_n\|^2 - \frac{p_n^\top P_{-n}}{N-1} - \frac{p_n^\top (x_n + X_{-n} - c)}{N-1} \right. \\ & \left. + \frac{NP_{-n}^\top (x_n - c/N)}{(N-1)^2} + \frac{S_{-n}}{2(N-1)} - \frac{D_{-n} - P_{-n}^\top c/N}{(N-1)^2} \right]. \end{aligned} \quad (2)$$

All derivatives below are taken from this expression.

2.2 Obstructions in Q

Proposition 1. *Q's strict certificate P1(ii) and implementation equality P2(i) cannot hold simultaneously. Moreover, payment (2) need not be individually rational, even with normalised strongly concave valuations.*

Proof. In Q's notation, P2(i) requires $\sum_m A_{nm}^n = 0$. For $v \neq 0$ and $h = \text{col}((0, v), \dots, (0, v))$, its price blocks give

$$h^\top (\Psi + \Psi^\top) h = -2 \sum_n v^\top \left(\sum_m A_{nm}^n \right) v = 0.$$

P1(ii), $\Psi + \Psi^\top \preceq -vI$ with $v > 0$, would instead bound this by $-vN\|v\|^2 < 0$. This disproves simultaneous feasibility of those sufficient conditions.

For the payment claim, take $N = 3$, $\gamma = c = 1$, $X_n = [0, 1]$, and valuations

$$V_1(x) = 10x - x^2/2, \quad V_2(x) = V_3(x) = 0.1x - x^2/2, \quad f_n = -V_n.$$

At $x = (1, 0, 0)$, $p = (9, 9, 9)$, own price derivatives vanish and own allocation cost derivatives are 0, 8.9, 8.9. The positive derivatives occur at lower bounds. Each own cost Hessian is $\begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix} \succ 0$, so these conditions suffice for a Nash equilibrium. Substitution gives $t_1^Q = 10.5 > V_1(1) = 9.5$. $\square$

More generally, put $b_N = (N^2 - N + 1)/(N - 1)^2$. At common prices $p_n = p$ satisfying $p^\top(\sum_n x_n - c) = 0$, collection of terms gives

$$t_n^Q = \gamma b_N p^\top (x_n - c/N). \quad (3)$$

The printed linear coefficients also satisfy $\sum_m a_m^n = \gamma c/[N(N - 1)]$, violating Q's P4(ii) when capacity has a positive component. These failures do not imply absence of equilibria or impossibility of every implementing mechanism.

2.3 A direct replacement theorem

Assume each f_n is continuous and m -strongly convex on X_n , with convex extensions sufficient for the subgradient and normal-cone sum rules. Assume a planner KKT pair exists. Define

$$t_n^{\text{corr}} = t_n^Q + \gamma(b_N - 1) \left(\frac{P_{-n}}{N - 1} \right)^\top \left(X_{-n} - \frac{N - 1}{N} c \right). \quad (4)$$

Theorem 2. *If*

$$m > \frac{\gamma(2N - 1)^2}{4(N - 1)^4}, \quad (5)$$

then every equilibrium of the corrected game implements the unique planner allocation. Prices are common and $\lambda = \gamma p_n$ is a planner multiplier. Conversely, every planner KKT pair gives an equilibrium at $p_n = \lambda/\gamma$. If also $0 \in X_n$ and $f_n(0) = 0$, then for $V_n = -f_n$ the corrected equilibrium payments satisfy

$$t_n^{\text{corr}} = \lambda^\top (x_n - c/N), \quad \sum_n t_n^{\text{corr}} = 0, \quad V_n(x_n) - t_n^{\text{corr}} \geq \lambda^\top c/N \geq 0. \quad (6)$$

Prices need not be unique.

Proof. The added term depends only on opponents, preserving every best response. Stack allocations and prices separately. Let $\Pi = \mathbf{1}\mathbf{1}^\top/N$, $\Delta = I - \Pi$, and $\kappa = (2N - 1)/(N - 1)^2$; these matrices act resourcewise. Differentiating (2) and multiplying price rows by $(N - 1)/N$ gives the weighted own-cost operator

$$G_x \in \partial f(x) + \gamma(\Pi - \kappa\Delta)p, \quad G_p = \gamma\Delta p - \gamma\Pi x + \frac{\gamma}{N}\mathbf{1} \otimes c. \quad (7)$$

Positive row scaling preserves the product-set optimality conditions. For $u = x - x'$, $v = p - p'$, strong convexity and completion of squares give

$$\begin{aligned} \langle G(s) - G(s'), s - s' \rangle &\geq m\|\Pi u\|^2 + (m - \gamma\kappa^2/4)\|\Delta u\|^2 \\ &\quad + \gamma\|\Delta v - (\kappa/2)\Delta u\|^2. \end{aligned} \quad (8)$$

This holds for any corresponding subgradient selections. Condition (5) also implies $m > \gamma/(N - 1)^2$, so the own allocation-price cost is jointly strongly convex: its price quadratic has curvature γ and cross coefficient $-\gamma/(N - 1)$. Thus own optimality conditions are sufficient.

Substituting a KKT pair into (7) supplies a game equilibrium. Add its variational inequality to that of any other equilibrium. The left side of (8) is then nonpositive, forcing $u = 0$ and $\Delta v = 0$. At common prices the remaining game conditions are precisely (1). Strong convexity gives allocation uniqueness.

Equation (3), the correction and complementarity yield the transfer identity and budget balance. Choose $g_n \in \partial f_n(x_n)$, $v_n \in N_{X_n}(x_n)$ with $g_n + \lambda + v_n = 0$. Feasibility of zero gives $v_n^\top x_n \geq 0$. The subgradient inequality at zero yields $0 \geq f_n(x_n) + \lambda^\top x_n$, proving participation. $\square$

3 The fixed risk surrogate and economic errors

Assume $0 \in X_n$. Losses $J_n(x, \xi)$ are bounded in absolute value by U_n , samplewise L_n -Lipschitz and m_n -strongly convex on X_n . Samples are fresh and iid from a query-independent distribution. For upper-tail probability $\beta_n \in (0, 1)$, define

$$C_n(x) = \inf_{\tau \in \mathbb{R}} \left\{ \tau + \beta_n^{-1} \mathbb{E}[J_n(x, \xi) - \tau]_+ \right\}, \quad (9)$$

$$\widehat{C}_{n,s_n}(x) = \inf_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{\beta_n s_n} \sum_{i=1}^{s_n} [J_n(x, \xi_i) - \tau]_+ \right\}. \quad (10)$$

Here $[a]_+ = \max\{a, 0\}$. True normalised valuation is $V_n(x) = C_n(0) - C_n(x)$. For satisfaction data the loss is $J_n = -\psi_n$; cash translation allows deterministic taxes outside the risk functional. The planner sums individual risks, rather than taking the risk of an aggregate random loss.

3.1 One-sided value error and feasible queries

Suppose a known relative interior ball $a_n + \rho_n \mathcal{B}_n \subset X_n$ is available, where $\mathcal{B}_n$ is the unit ball in the local affine hull, of dimension $d_n > 0$. Fix $0 < \varepsilon_n < 1$ and $0 < \delta_n < \varepsilon_n \rho_n$. Define

$$q_n(x, v) = (1 - \varepsilon_n)x + \varepsilon_n a_n + \delta_n v, \quad (11)$$

$$K_n(x) = \mathbb{E}_{v, \xi^{1:s_n}} \widehat{C}_{n,s_n}(q_n(x, v)), \quad v \sim \text{Unif}(\mathcal{B}_n). \quad (12)$$

Decisions remain in X_n , including zero. Set

$$e_n = \frac{U_n}{\beta_n} \sqrt{\frac{2\pi}{s_n}}, \quad R_n = \sup_{x \in X_n} \|x - a_n\|, \quad w_n = e_n + 2L_n \varepsilon_n R_n + L_n \delta_n. \quad (13)$$

Lemma 3. *Every query in (12) is locally feasible, and*

$$-e_n - L_n \varepsilon_n R_n \leq K_n(x) - C_n(x) \leq L_n \delta_n + L_n \varepsilon_n R_n. \quad (14)$$

The surrogate has curvature $m_n(1 - \varepsilon_n)^2$. With a uniform sphere direction u , independent of the fresh batch, the oracle

$$\widehat{g}_n(x) = \frac{(1 - \varepsilon_n)d_n}{\delta_n} \widehat{C}_{n,s_n}(q_n(x, u))u \quad (15)$$

is unbiased for $\nabla K_n(x)$, the relative gradient. It satisfies

$$\|\hat{g}_n(x)\| \leq (1 - \varepsilon_n)d_n U_n / \delta_n.$$

Proof. The query is the convex combination $(1 - \varepsilon_n)x + \varepsilon_n(a_n + (\delta_n/\varepsilon_n)v)$. The strict radius inequality also leaves a relative neighbourhood on which the smoothing identity is valid. Applying P's sphere identity and the chain rule gives (15). Since $0 \in X_n$, the affine hull is a linear subspace; gradients can be lifted by an orthonormal basis.

Let $H_n = \mathbb{E}\hat{C}_{n,s_n}$. Interchanging infimum and expectation in the inequality direction gives $H_n \leq C_n$. P's integrated Dvoretzky–Kiefer–Wolfowitz calculation gives $\mathbb{E}|\hat{C}_{n,s_n}(x) - C_n(x)| \leq e_n$, hence $C_n - e_n \leq H_n \leq C_n$. At a centre y admitting ball queries, Jensen's inequality and Lipschitz continuity therefore give

$$-e_n \leq \mathbb{E}_v H_n(y + \delta_n v) - C_n(y) \leq L_n \delta_n. \quad (16)$$

Taking $y = (1 - \varepsilon_n)x + \varepsilon_n a_n$ proves (14). These compare deterministic expected functions; no expected supremum of random empirical errors is used.

Samplewise strong convexity passes through CVaR by monotonicity, convexity and cash translation: the deterministic quadratic deficit passes through the risk functional unchanged. This applies also to empirical CVaR and expectation. Affine contraction scales the deficit by $(1 - \varepsilon_n)^2$. $\square$

3.2 Economic consequence

For a real function a , let $\text{osc}_X a = \sup_X a - \inf_X a$. Equation (14) gives $\text{osc}_{X_n}(K_n - C_n) \leq w_n$. Use surrogate valuation $V_n^K(x) = K_n(0) - K_n(x)$, and retain payment (4).

Theorem 4. *Assume a surrogate planner KKT pair exists and*

$$m_{\text{eff}} = \min_n m_n (1 - \varepsilon_n)^2 > \frac{\gamma(2N - 1)^2}{4(N - 1)^4}.$$

At every surrogate equilibrium $\tilde{s}$, its allocation $\tilde{x}$ uniquely minimises $\sum_n K_n$ on $\mathcal{A}$. Budget balance and surrogate individual rationality are exact. Writing $U_n(s) = V_n(x_n) - t_n^{\text{corr}}(s)$, population-risk guarantees are

$$\sup_{s'_n \in X_n \times \mathbb{R}_+^r} \{U_n(s'_n, \tilde{s}_{-n}) - U_n(\tilde{s})\} \leq w_n, \quad (17)$$

$$U_n(\tilde{s}) \geq -w_n, \quad (18)$$

$$\sum_n C_n(\tilde{x}_n) - \sum_n C_n(x_n^*) \leq \sum_n w_n, \quad (19)$$

where x^ minimises population cost on $\mathcal{A}$. If total population cost has curvature $m_0 > 0$, then*

$$\|\tilde{x} - x^*\|^2 \leq \frac{2}{m_0} \sum_n w_n. \quad (20)$$

Proof. Lemma 3 and Theorem 2 give exact surrogate implementation and transfers. Put $a_n = K_n - C_n$ in this proof only. The true minus surrogate payoff at allocation x_n is $a_n(x_n) - a_n(0)$. Its change under any unilateral deviation is at most $\text{osc } a_n$, proving (17); its absolute value is at most w_n , proving (18). Transfers themselves do not change when evaluated under true preferences, so budget balance remains exact. Surrogate optimality gives

$$\sum_n [C_n(\tilde{x}_n) - C_n(x_n^*)] \leq \sum_n [a_n(x_n^*) - a_n(\tilde{x}_n)] \leq \sum_n w_n.$$

Strong convexity and constrained optimality at x^* lower-bound this gap by $(m_0/2)\|\tilde{x} - x^*\|^2$. $\square$

Taking $\delta_n = \varepsilon_n \rho_n / 2$ yields error order $O(s_n^{-1/2} + \varepsilon_n)$, for fixed problem constants. Where centred queries are already permissible, (16) gives the smaller oscillation bound $e_n + L_n \delta_n$. The improvement is at the objective level and does not remove the inverse-radius dependence of oracle variance.

4 A replacement learner

Fix all surrogate parameters. Assume a compact price box $[0, P_{\max}]^r$ containing a surrogate KKT equilibrium, and put $\mathcal{S} = \prod_n (X_n \times [0, P_{\max}]^r)$. Comparison with that equilibrium in (8) shows every boxed equilibrium has its allocation and common prices. Capacity is therefore feasible already; slack resources force zero price by the boxed price condition. No strict interiority of the reference price is needed. This retains the unboxed equilibrium characterisation.

Let G^K denote (7) for the smooth costs $K_n - K_n(0)$. For $\eta > 0$, define

$$R(z) = (I + \eta(G^K + N_{\mathcal{S}}))^{-1}z, \quad H_z(s) = G^K(s) + \eta^{-1}(s - z).$$

The continuous monotone operator on the compact convex product gives a well-defined resolvent. To approximate it, use projected stochastic updates with the oracle (15), analytic payment gradients, and aggregate allocation and price messages:

$$s_{t+1} = \text{proj}_{\mathcal{S}}(s_t - \alpha_t \hat{H}_z(s_t)), \quad \alpha_t = \frac{1}{\sigma(t+2)}, \quad \sigma = \eta^{-1}.$$

Proposition 5. *Run $T_k = \lceil (k+1)^{2+\zeta} \rceil$ inner updates centred at z_k , where $\zeta > 0$, and set $z_{k+1} = s_{T_k}$. Then z_k converges almost surely and in mean square to a possibly random surrogate equilibrium. Its allocation converges in mean square to the deterministic optimum $\tilde{x}$ of Theorem 4.*

Proof. Compactness and the oracle bound give a uniform conditional second moment M^2 for $\hat{H}_z$. With $y = R(z)$, projection, the VI at y , and σ -strong monotonicity give

$$\mathbb{E}[\|s_{t+1} - y\|^2 \mid s_t, z] \leq (1 - 2\sigma\alpha_t)\|s_t - y\|^2 + \alpha_t^2 M^2.$$

Induction yields

$$\mathbb{E}[\|s_T - R(z)\|^2 \mid z] \leq C/(T+1), \quad C = \max\{\text{diam}(\mathcal{S})^2, M^2/\sigma^2\}.$$

Thus the errors $e_k = z_{k+1} - R(z_k)$ have $\sum_k \mathbb{E}\|e_k\| < \infty$, and are almost surely summable.

Subtracting the resolvent VIs proves firm nonexpansiveness. For any fixed point z^* ,

$$\|R(z_k) - z^*\|^2 \leq \|z_k - z^*\|^2 - \|z_k - R(z_k)\|^2.$$

Summable errors imply convergent distances to fixed points and vanishing residual. Compactness supplies a cluster point; continuity places it in the fixed-point set. The distance argument then gives convergence of the whole sequence. Boundedness yields mean-square convergence by dominated convergence. The allocation component is unique by Theorem 2. $\square$

5 Related work and interpretation

The sampled CVaR one-point step is already published. Wang, Shen and Zavlano [5], Algorithm 1 and Lemmas 2–3, give the estimator, smoothing comparison and distribution-function error bound, with regret guarantees. The nested learner is also an application of an existing principle: Cui and Shanbhag [2], Section 3.2, Propositions 5–6, use stochastic inner solves of proximal inclusions and a polynomial inner schedule of exponent greater than two to obtain almost-sure convergence under monotonicity. We include the compact-case proof to specify precisely which learner is justified here.

Wang et al. [6], Algorithm 1 and Theorem 1, use loss gradients and estimated quantiles under strong monotonicity. Their theorem bounds the time average of squared distance with high probability. Our weighted operator retains a common-price degeneracy. Cherukuri [1], Section 3, studies sample-average VIs whose operator components are CVaRs, proving consistency and exponential solution-set approximation. Here the implemented objective is the deterministic expectation of a fixed empirical risk estimate, and the operator differentiates that objective. Those statistical targets and guarantees differ.

The comparison supports a focused contribution: a correction of Q’s specific payment, with P’s fixed surrogate interpreted through explicit welfare and incentive errors. It does not support novelty claims for CVaR game learning, the proximal construction or feasible smoothing in general. The sources examined did not identify the same payment repair and economic-bound package; this bounded search is not a proof of priority. Economically, finite sampling changes the implemented preferences. Equations (17)–(19) concern a fixed surrogate equilibrium assessed using population risk, rather than a finite-time regret guarantee for a learning trajectory.

6 Limitations and open questions

The added assumptions are substantive: samplewise curvature, valid KKT sum rules, existence of a surrogate planner multiplier, a feasible zero outside action, known local interior balls and fresh losses at arbitrary local queries. For learning, a known price cap must contain a KKT price. Q’s expected curvature assumption alone does not imply CVaR curvature. Exact budget balance is asserted only at equilibrium; local query feasibility does not ensure joint capacity feasibility of experiments. No efficient total oracle complexity or stochastic benchmark is supplied.

Weighted monotonicity does not prove Q’s proximal best-response map non-expansive. In the recorded interior quadratic example with $N = 3$ and $m = \gamma = 1$, its consensus-price column has squared norm $1 + 1/(2D^2) > 1$, where $D = (\mu + 1)^2 - 1/4$, for every $\mu > 0$. This defeats that certificate, without proving divergence of all damped iterations. Proposition 5 concerns a changed algorithm.

Allocation uniqueness likewise does not imply price uniqueness. For $N = 3$, $X_n = [0, 1]$, $c = 3/2$, $\gamma = 1$, equiprobable $Z = \pm 1$, and tail probability $1/2$, the losses

$$J_n(x, Z) = x^2 - 3x + Z(x - 1/2)$$

give $C_n(x) = x^2 - 3x + |x - 1/2|$. The unique allocation is $x_n = 1/2$, but $\partial C_n(1/2) = [-3, -1]$ admits every common price in $[1, 3]$.

Open questions are convergence of the original VS-PBR iteration, efficient sample complexity for the replacement learner, a population-risk limit with changing batches and radii, and jointly capacity-feasible exploration. The fixed-parameter theorem establishes none of these. Broader publication priority of the specific correction remains subject to further comparison.

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